Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Understanding Cases 6.0.0

Learn more at Stellar Cyber Academy.

The following links take you to courses on the Stellar Cyber Academy technical training portal where you can learn more about a topic by watching the suggested lessons.

J02 Introduction to Features in the User Interface (02h:24m)

Case Management (48m:00s)

Watch a demonstration of Case Management workflows in the Platform, covering how to interact with, filter, and sort cases using the Top Cases view and toolbar filters. Explore the search tool and other filtering options that support efficient investigation workflows across the data lake. Follow along as a case is investigated using detailed analysis options and visualizations, tracking the progression of an attack through kill chain stages. Learn how machine learning and threat intelligence surface relevant Interflow records and contextual insights to help analysts understand and respond to security incidents.

J03 Investigations and Workflows (05h:48m)

Workflow Tools in the Stellar Cyber UI (35m:00s)

Explore workflow tools within the UI, including case management, knowledge sharing, task assignment, and alert triage. Discover ways to enhance collaboration, automate notifications, and integrate the UI tools into efficient SOC workflows.

Built-In Investigation Views (25m:00s)

Get familiar with built-in investigation views within the Platform, such as case views, alerts, and kill chain views. Learn how to prioritize and sort cases and alerts for effective triage, using these views to drive incident response.

Building Workflows in Stellar Cyber - In Practice (18m:00s)

Dive into practical considerations for creating workflows, including best practices for workflow consistency and efficiency. Explore tips for building workflows that adapt to real-world security operations and SOC needs.

Case Investigation (45m:00s)

Watch a step-by-step investigation of a case in the Platform from start to finish. Follow an example case through detailed analysis and evidence gathering, learning how to apply tools and techniques — including toolbar filters, the search tool, and threat intelligence — to build a cohesive incident narrative. Explore how to enrich cases with context drawn from Interflow data lake records and different stages in the kill chain for a more comprehensive investigation using Case Management.

The first time you access a link on the portal during a session, you must log in to access content.

Stellar Cyber leverages ML to correlate disparate alerts into coalesced *cases*.

A case represents a grouping of potentially related alerts in a single data structure. Cases provide holistic context, allowing the analyst to examine the case and its associated alerts to assess whether the case represents a real attack, true high-risk behavior, or event connections without security significance.

To understand why the alerts in a case are potentially related, consider the following example: a trojan alert and a traffic anomaly alert found on the same asset within an hour of one another. This could represent real malware or it could just be a coincidence. Additional related alerts and scoring will help distinguish the difference. The analyst can work with the algorithm collaboratively to apply context and investigate what is potentially threatening or risky.

Cases accelerate complex detection and response by providing a higher-level construct to analyze instead of individual alerts. Compared to the triage of individual alerts, a case provides more context of correlated behaviors to help analysts make a more holistic evaluation during triage.

Refer to the following sections for details:

- **How Case Correlation Works**
    - **About the Correlation Timeout**
    - **How Connection Strength Works**
    - **Case Correlation Details**
    - **Case Correlation and Manually Generated Events**
- **Customizable Case Correlation Strategies**
    - **Selecting and Evaluating a Case Correlation Strategy**
    - **About Multi-Entity Case Correlation**
- **How Case Scores Are Calculated**
    - **Score Calculation Details**
- **Cases in the Stellar Cyber User Interface**

How Case Correlation Works 5.5.0

Stellar Cyber uses a risk-centric algorithm for case correlation by default. The risk-centric algorithm works as follows.

You can experiment with alternative case correlation strategies as part of the Early Access Program.

1. Stellar Cyber generates an alert.
2. The case correlation algorithm attempts to associate the alert with a case in real time.
    - If there is a strong connection with an existing case, the alert is grouped and correlated with that case. One alert can belong to multiple cases; the algorithm attaches the alert to any case as long as there is a strong enough connection.

      For system-created cases, the Stellar Cyber Platform does not create a case activity entry when an alert is correlated and added to an existing case. Instead, it records changes to the case activity history when the case score, which indicates how critical a case is, increases or decreases by 3 points or more.

      For user-created cases, the case activity history tracks alerts being added to or removed from the case.
    - If there is no strong connection with an existing case, what happens next depends on whether Case Filters are enabled in your deployment:
        - **Case Filters Not Enabled** – A new case is created containing this alert. The result is a new case with this single alert.

          This is the same behavior as in previous Stellar Cyber releases.
        - **Case Filters Enabled** – A case candidate (sometimes referred to as a "proto-case") is created with this single alert. The case candidate is evaluated against the current Case Filter settings. 5.5.0
            - If the candidate matches any of the case filters, the case is suppressed and is not displayed. It remains a candidate until it meets one of the thresholds in the next step. Until the candidate ages out according to one of the thresholds in the next step, it can still become a full case if filter criteria change.
            - If the candidate does not match any of the case filters, it is promoted to a full case and is displayed in the Cases list.
3. A Case continues to accept new correlated alerts and grow until whichever of the following happens first:
    - The case is closed by a user (**Status** is set to either **Resolved** or **Cancelled**).
    - The number of alerts associated with the case reaches the maximum specified by the **Alerts Per Case** threshold (5,000, by default). 6.2.0

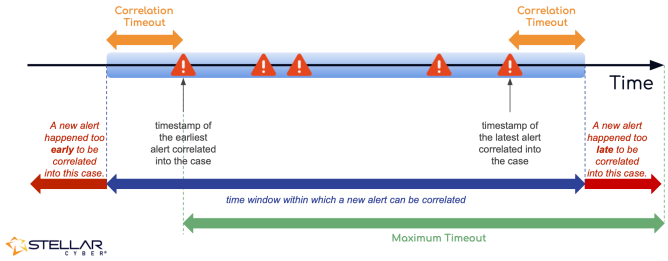
- No new alert is associated with the case within the time window specified by the global Correlation Timeout.
- The duration of a case exceeds 30 days.

This means that the maximum duration for a case is 30 days (the time between the first and last alerts) as long as new alerts come in more frequently than the Correlation Timeout and the case is not closed by a user.

4. Alerts with a status of **Closed** or **Ignored** are not correlated to cases.

About the Correlation Timeout

An alert can be correlated into an existing case if it occurs within a specific time window of the case. This time window is determined by **Correlation Timeout**, which is the amount of time that has passed *after* the latest alert that was correlated into the case or *before* its earliest correlated alert. The figure below illustrates how this works:



As summarized in the figure above:

- A new alert is not considered for correlation if it occurs *after* the amount of time specified by the **Correlation Timeout** has passed since the last correlated alert for the case. An alert such as this has occurred **too late** for correlation.
- A new alert is not considered for correlation if it occurs earlier than the amount of time specified by the **Correlation Timeout** before the earliest alert associated with the case. An alert such as this has occurred **too early** for correlation into the case.

The default **Correlation Timeout** is three hours; the maximum is 24 hours.

Refer to **Setting the Correlation Timeout** for details on configuring a global correlation window for your organization.

How Connection Strength Works

The connection strength between an alert and a potential case has to do with shared entities (for example, assets or users), shared properties (for example, hashes or URLs) and time (close time windows), which are also considered to evaluate how to build strong context. The algorithm is trained on real-world attacks and data; that training is what informs the level of connection strength necessary for correlation. The algorithm continues to improve as more data is incorporated into training, thereby improving case output as well.

Case Correlation Details

Stellar Cyber correlates both asset and user-based alerts into cases. In general, if an alert has any of the fields in the tables below populated, it is correlated into a case:

Asset-Related Fields for Case Correlation

Alerts that have any of the following asset-related fields populated are correlated into cases:

Category	Fields
Asset ID	dstip_assetid hostip_assetid srcip_assetid
IP	dstip host.ip hostip srcip IP-based correlation is performed for internal IP addresses and not public IP addresses. If no internal IP addresses are found as part of an alert, Stellar Cyber uses other fields for correlation (for example, the user fields listed in the table below). Stellar Cyber uses private IP addresses for case correlation to track activities and identify compromised systems within the boundaries of an organization, allowing incident responders to focus on assets and endpoints under their jurisdiction. This approach also excludes external IP addresses that change frequently and can be shared across multiple users, (for example, CDN or proxy servers), making attribution unreliable and potentially leading to inaccurate or incomplete results. Stellar Cyber does, however, use external IP addresses for threat intelligence and geolocation tracking.

Host Name	`computer_name`
	`dstip_host`
	`host.name`
	`hostip_host`
	`srcip_host`

Cloud Resource ID	cloud.resource.id Stellar Cyber can correlate alerts to cases based on matches in the Cloud Resource ID using data from integrations with Microsoft Defender for Cloud, AWS GuardDuty, AWS CloudTrail (SigmaHQ rules), and Azure Activity Logs (SigmaHQ rules). You may see alerts correlated to cases from other cloud sources (for example, OCI CloudGuard), but those will be based on matches in other fields (IP addresses and user names, for example) rather than Cloud Resource IDs.

User-Related Fields for Case Correlation 6.3.0

Alerts with any of the following user-related fields populated are correlated into cases:

Category	Fields
User SID	dstip_usersid event_data.SubjectUserSid event_data.TargetUserSid hostip_usersid srcip_usersid
User ID	user.id
Username	dst_name dst_user 6.5.0 dstip_username event_data.SubjectUserName event_data.TargetUserName hostip_username principal_name 6.5.0 secondary src_user 6.5.0 srcip_username user.name username wineventlog_user.name
Email	email.message_id (when msg_origin is either proofpoint_tap or mimecast_email) user.email

6.3.0 Case correlation uses advanced user identity matching to associate alerts that reference the same individual, even when different data sources provide varying or incomplete user identifiers. This ensures that alerts linked to a single user are grouped into one correlated case, giving you a unified view of user activity and associated risks.

The correlation engine evaluates multiple identifiers in a prioritized sequence to determine whether alerts refer to the same user. When the `user.email` and `user.name` identifiers overlap across data sources, Stellar Cyber merges the alerts into a single user entity and groups them into one case. This approach consolidates investigations that span endpoints, networks, and cloud services, allowing you to see a complete picture of a user's behavior across environments.

The correlation engine also uses this normalized user identity when constructing correlation graphs. 6.5.0 As a result, detections, timelines, and pivots can reference a single user node across multiple data sources even when the user appears under different fields such as

`user.name`, `user.id`, `principal_name`, `src_user`, or `dst_user`. This reduces fragmented user nodes and helps preserve relationships that might otherwise be missed during investigation.

User identity correlation extends to Microsoft security products, including Microsoft Defender, Azure Active Directory Risk Detection, and Office 365 Audit for Azure Active Directory. For these integrations, Stellar Cyber normalizes user identifiers such as `user.name`, `username`, `user.email`. It also converts all email usernames to lowercase before evaluation to ensure consistent matching across data sources. Variations in capitalization—such as `John.Smith@Company.com` and `john.smith@company.com`—are treated as the same user.

Example – A user named John Smith appears in different alerts from Microsoft services using several identifiers:

- In Microsoft Defender, as `user.name: John.Smith`
- In Azure Active Directory Risk Detection, as `username: john.smith@company.com`
- In Office 365 Audit for Azure Active Directory, as `user.email: JOHN.SMITH@COMPANY.COM`

Although the identifiers differ in format and capitalization, case correlation recognizes them as belonging to the same individual. The alerts are merged into a single case, allowing you to view all related activity for John Smith in one place and assess whether it represents a coordinated compromise attempt.



For alerts from `office365_audit_azureactivedirectory` and `azure_ad_risk_detection`, case correlation prioritizes user identity over host identity. This ensures that alerts referencing the same user are grouped together, even when host information differs.

Correlation for Alerts with Multiple Observables 6.1.0

Many Stellar Cyber alerts have multiple associated observables. Alerts with multiple observables are sometimes referred to as *summarized* alerts.

Case Management's ability to correlate and display multiple observables from summarized alerts is described in the table below:

Summarized Alert	Correlation/Display Support
Internal IP Port Scan Anomaly	Display of multiple destination IP addresses as case observables.
Internal SMB Read Anomaly	Display of multiple SMB paths as case observables.
Internal SMB Write Anomaly	
Internal Credential Stuffing	Correlation and display of suspicious IP addresses in alerts.
External Credential Stuffing	
Internal Password Spraying	Correlation and display of target users in alerts.
External Password Spraying	
Password Resets Anomaly	Correlation and display of target users in alerts.

Summarized Alert	Correlation/Display Support
Account Creation Anomaly	Correlation and display of users in alerts.
Cloud Drive Data Exfiltration Anomaly	Display of multiple files as case observables.

Alerts Correlated to Cases

Stellar Cyber creates cases for **all** alerts, including those without any observables. Only the following alerts are not correlated into cases:

- Alerts from the Sensor index (**Data Ingestion Anomaly** and **Sensor Status Anomaly**).
- Alerts from correlation-based ATH rules.

This means that Stellar Cyber creates single-alert cases for the following situations:

- Alerts where no user or asset can be found.
- Alerts from third-party cloud alert integrations, such as AWS GuardDuty.
- Alerts from automated threat hunting with only raw data.



All alerts that can be correlated to cases can also be synchronized to ServiceNow via an active InSync, when configured.

Searching for Alerts Without Correlated Cases

You can use the following search in the Lucene Search bar in either the Alerts or Threat Hunting page to see alerts that have not been correlated to cases in the current time window:

```
(NOT ( (_exists_:(xdr_event.name OR event_name) AND _exists_:orig_index) OR
(stellar.ath.type:raw AND stellar.ath.to_incident:true))) OR xdr_event.name:
(ade_outbytes_anomaly OR ade_outbytes_anomaly_flip)
```

Cases with Cloud Observables from Third-Party Alert Integrations

Stellar Cyber identifies and presents cloud observables for cases based on alerts from the following third-party cloud integrations:

- Microsoft Defender for Cloud
- AWS GuardDuty
- SigmaHQ rules for AWS CloudTrail
- SigmaHQ rules for Azure Activity Logs

In addition, Stellar Cyber correlates alerts to cases using matching Cloud Resource IDs from Microsoft Defender for Cloud, AWS GuardDuty, AWS CloudTrail, and Azure Activity Log alerts, as described in **Case Correlation Details**.

Cases with Observables from Cloud-Based Email Integrations

Stellar Cyber identifies and presents observables for cases based on alerts from the following third-party email integrations:

- Mimecast
- Proofpoint

Refer to **Observables from Cloud-Based Email Security Integrations** for information on the observables Stellar Cyber extracts and displays in the Case Analysis workspace based on email alerts from these sources.

Case observables from cloud-based email integrations are available as part of an Early Access Program and might not be available in your version of the Stellar Cyber Platform.

Case Correlation and Manually Generated Events

Considerations for manual event generation: Due to the nature of Stellar Cyber's alert correlation and machine learning models, manually generated or ad-hoc events may not always trigger the expected alert or case creation. The Stellar Cyber platform is designed to analyze and correlate multiple events over time, assessing patterns and context to determine the severity of a potential threat. As a result, isolated or artificial events might not meet the criteria for high-severity alerts or case generation, and may not be representative of the system's full detection capabilities in real-world threat scenarios.

Customizable Case Correlation Strategies 6.0.0

This section describes a feature that is currently only available as part of an Early Access Program and may not be enabled in your version of the Stellar Cyber Platform. Contact your sales representative for information on enabling this feature and comparing its results to those of the traditional risk-centric model.

Stellar Cyber provides support for multiple case correlation strategies as an EAP feature, allowing teams to evaluate and experiment with different approaches to grouping alerts into cases. Each strategy provides a distinct investigative perspective:

- Multi-Entity Correlation links alerts across interconnected hosts and actions to form a single case, offering a holistic view of extended or lateral attack campaigns.
- Attacker-Centric Correlation groups alerts by the source (attacker) host, making it easier to track adversary behavior across multiple targets.
- Victim-Centric Correlation organizes alerts by the destination (victim) host, enabling focused protection and visibility on high-value assets.

The flexibility to choose a case correlation strategy enables security teams to tailor investigations based on their operational priorities, whether that's identifying persistently targeted endpoints, tracing threat actor movements, or capturing full-scale intrusion campaigns. To join the EAP and begin testing these correlation strategies, contact your Stellar Cyber Customer Success representative.

Selecting and Evaluating a Case Correlation Strategy

You must work with Stellar Cyber Customer Success to change the case correlation strategy for your deployment – you can't change it by yourself. Regardless of the strategy you select, plan on allowing at least a week to evaluate a given case correlation strategy. During the evaluation

period, monitor your deployment for sudden changes in case graphs or volumes. In addition, pay attention to the following key metrics to determine which strategy is best for you:

Metric	Why it Matters	Primary Case Correlation Strategy
Number of Cases Created	A sharp drop signals successful consolidation of alerts. This is especially desirable when evaluating the Multi-Entity strategy.	Multi-Entity
Average Alerts per Case	Higher averages indicate stronger grouping of alerts, which is crucial for the Multi-Entity strategy.	Multi-Entity
Threat-Actor Isolation Rate	The percentage of cases tied to a single source host can measure how well case correlation is locking onto one attacker.	Attacker-Centric
Asset-Impact Grouping	The percentage of alerts per destination host grouped together shows the effectiveness of Victim-Centric mode.	Victim-Centric
Case Resolution Time	Faster turnaround of cases reflects overall workflow efficiency. This is a useful barometer for comparing all strategies.	All
Analyst Confidence (qualitative)	Direct feedback on clarity and usability, which is valuable across every correlation strategy.	All
False-Negative / Missed Events	Track any gaps where alerts should have been grouped but weren't. This is important for detection quality.	All
True Positive Ratio per Analyst	Monitor the proportion of alerts or cases each analyst confirms as true positives. This can help you understand how alert grouping affects accuracy.	All

About Multi-Entity Case Correlation

This section describes a feature that is currently only available as part of an Early Access Program and may not be enabled in your version of the Stellar Cyber Platform. Contact your sales representative for information on enabling this feature and comparing its results to those of the traditional risk-centric model.

This release includes *multi-entity case correlation* as an Early Access Program feature, allowing a given alert to be correlated to cases based both on its associated assets (hosts) and its associated users. This is a contrast to the current single-entity model that identifies the most important asset or user for an alert and correlates it to cases based on that entity. The table below summarizes the differences in the two approaches:

Single-Entity Correlation	Multi-Entity Correlation
For each new alert, Stellar Cyber: 1. Identifies all hosts and users (entities) associated with the alert. 2. Chooses one of the associated entities as the most essential for correlation based on an assessment of its risk. This is the entity that is most concerning from a security perspective and is more often a host than a user because of its relative importance as a vector for attacks (for example, a host used in a port scan). 3. Checks for other active cases that involve the same entity.	When multi-entity correlation is enabled, Stellar Cyber maintains an internal **Correlation Entities** list for each case. This list contains all of the hosts and users associated with the alerts in the case. Then, for each new alert Stellar Cyber: 1. Identifies all hosts and users (entities) associated with the alert.

Single-Entity Correlation	Multi-Entity Correlation
• If a match is found, associates the alert with those cases. • If no match is found, creates a new case based on the entity.	2. Checks for other active cases that have a matching host or user in its **Correlation Entities** list. • If a match is found, the alert is added to the case. In addition, all other hosts and users associated with the alert are added to the case's **Correlation Entities** list. • If no matches are found, creates a new case for the alert and adds all of its hosts and users to the case's **Correlation Entities** list.

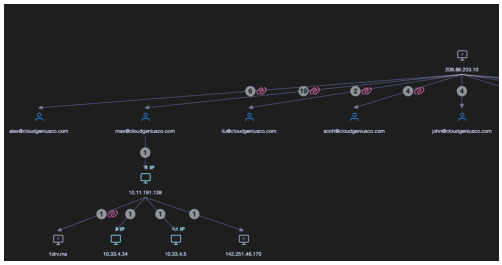
Multi-Entity Case Correlation in Action

Multi-entity case correlation enables Stellar Cyber to correlate alerts to cases based on matches in either a host or a user. For example, consider the following alerts:

- **Alert 1** – Involves user-x
- **Alert 2** – Involves user-x and host-a
- **Alert 3** – Involves host-a and host-b

In the single-entity model, alerts such as these would result in separate cases. The multi-entity model, however, can create a single case that includes all of the alerts. The intent of this approach is to capture the entire path of an attack, including the interactions between hosts and users. For example, multi-entity correlation can give you a coalesced view of a phishing attack that results in a compromised host, displaying both the affected hosts and users in a single case.

The figure below provides an example of multi-entity correlation in action, with both hosts and users in the same case graph:



How Case Scores Are Calculated

Stellar Cyber assigns scores to cases based on how critical they are. The score updates in real time as events and entities are added to or removed from the case. This section explains how case scores are calculated.

In general, the score of a case is determined by the number of different alert types associated with it. The score typically increases as the number of distinct alert types associated with it also increases.

In addition to alert diversity, case scores are influenced by the characteristics of the associated alerts. In particular, the fidelity and severity of alerts have a significant impact on the calculation of a case score:

- **Fidelity** – The confidence in the analysis. Higher fidelity indicates greater confidence that a malicious event was correctly identified and increases the alert score. Lower fidelity, especially when combined with low Threat Intel scores, reduces the alert score.

- **Severity** – The importance of the category of the event. Higher severity reflects greater potential impact. In general, later-stage events have higher severity.

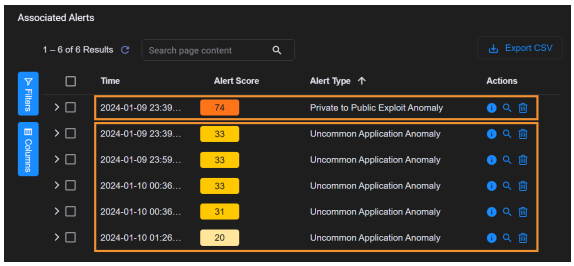
Case severity levels correspond to the following case score ranges:

Case Severity	Case Score
Critical	75 and higher
High	50–74
Medium	25–49
Low	1–24

If an alert filter adjusts an alert score, the updated value is included in the case score calculation. Open cases recalculate automatically to reflect the updated alert score.

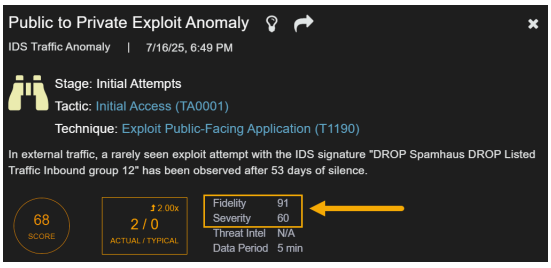
Score Calculation Details

Stellar Cyber begins to assign a case score by calculating an **Event Score** for each different alert type associated with the case. Stellar Cyber does this by summing the maximum Alert Score, Severity, and Fidelity for all individual alerts of a given type. For example, consider the case illustrated below:



This case has only two different alert types associated with it: **Private to Public Exploit Anomaly** and **Uncommon Application Anomaly**.

- There is only one **Private to Public Exploit Anomaly** alert, so Stellar Cyber uses the only Alert Score available, which is **74**. Similarly, it also uses the Severity and Fidelity scores from this alert. You can't see those in the table, but you can by selecting the **More Info** button in the table for the alert. For example:



- There are five different **Uncommon Application Anomaly** alerts associated with the case. As you can see in the table, the highest Alert Score across those five alerts is **33**, which is the value Stellar Cyber uses to calculate the score. Similarly, it also takes the highest Severity and Fidelity score across these five alerts.

Once done, there are separate Event Scores for both the **Private to Public Exploit Anomaly** and **Uncommon Application Anomaly** alert types. To arrive at a final score for the case, Stellar Cyber performs the following steps:

- Calculates a combined total score using the Event Scores for all alert types associated with the case. Because of this, the more different alert types that are associated with a case, the higher the score for the case will generally be.
- Normalizes the total score to a final score using the following equation: `final_score = (total_score * total_score) / 100.0`

Cases in the Stellar Cyber User Interface

Stellar Cyber reports cases in the XDR Kill Chain dashboard, as well as in the Cases interface, giving you a powerful tool to understand and respond to ongoing attacks.



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